

SOLIDWORKS API (.NET, VBA, C#) — Reference (208/337): IModelView (part 2 of 2)

SOLIDWORKS API Documentation (converted from CHM)

IModelView (part 2 of 2) (Type)

RemovePerspective Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : RemovePerspective Method (IModelView)

Removes perspective from the model view.

.NET Syntax

Visual Basic (Declaration)

```
Sub RemovePerspective()
```

Visual Basic (Usage)



```
Dim instance As IModelView  
instance.RemovePerspective()
```

C#

```
void RemovePerspective()
```

C++/CLI

```
void RemovePerspective();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::RemovePerspective](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::IGetEyePoint Method](#)
[IModelView::ISetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)
[IModelView::ObjectSizesAway Property](#)
[IModelView::ObjectSizesAway Property](#)

RollBy Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : RollBy Method (IModelView)

Rolls the model view about the line of sight by the specified angle.

.NET Syntax

Visual Basic (Declaration)

```
Sub RollBy(  
    ByVal Angle As System.Double  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Angle As System.Double
```

```
instance.RollBy(Angle)
```

C#

```
void RollBy(  
    System.double Angle  
)
```

C++/CLI

```
void RollBy(  
&    System.double Angle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle

Angle by which to roll the model view about the line of sight

Visual Basic for Applications (VBA) Syntax

See [ModelView::RollBy](#).

Example

[Roll Model View \(VBA\)](#)

See Also

[IModelView Interface](#)

[IModelView Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateAboutAxis Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : RotateAboutAxis Method (IModelView)

Rotates the model view about a point, by an angle in the specified direction.

.NET Syntax

Visual Basic (Declaration)

```
Sub RotateAboutAxis( _  
    ByVal Angle As System.Double, _  
    ByVal Ptx As System.Double, _  
    ByVal Pty As System.Double, _  
    ByVal Ptz As System.Double, _  
    ByVal AxisVecX As System.Double, _  
    ByVal AxisVecY As System.Double, _  
    ByVal AxisVecZ As System.Double _  
)
```

Visual Basic (Usage)

```
Dim instance As IModelView  
Dim Angle As System.Double  
Dim Ptx As System.Double  
Dim Pty As System.Double  
Dim Ptz As System.Double  
Dim AxisVecX As System.Double
```



```
Dim AxisVecY As System.Double
Dim AxisVecZ As System.Double
```

```
instance.RotateAboutAxis(Angle, Ptx, Pty, Ptz, AxisVecX, AxisVecY, AxisVecZ)
```

C#

```
void RotateAboutAxis(
    System.double Angle,
    System.double Ptx,
    System.double Pty,
    System.double Ptz,
    System.double AxisVecX,
    System.double AxisVecY,
    System.double AxisVecZ
)
```

C++/CLI

```
void RotateAboutAxis(
    & System.double Angle,
    & System.double Ptx,
    & System.double Pty,
    & System.double Ptz,
    & System.double AxisVecX,
    & System.double AxisVecY,
    & System.double AxisVecZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Angle
Angle of rotation
Ptx
Center of rotation
Pty
Center of rotation
Ptz
Center of rotation
AxisVecX
Direction of axis of rotation
AxisVecY
Direction of axis of rotation
AxisVecZ
Direction of axis of rotation

Visual Basic for Applications (VBA) Syntax

See [ModelView::RotateAboutAxis](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::RotateAboutCenter Method](#)
[IModelView::RotateAboutPoint Method](#)
[IModelView::StartDynamics Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateAboutCenter Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : RotateAboutCenter Method (IModelView)

Rotates the model view about the screen X and Y axes.

.NET Syntax

Visual Basic (Declaration)

```
Sub RotateAboutCenter( _  
    ByVal XAngle As System.Double, _  
    ByVal YAngle As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim XAngle As System.Double  
Dim YAngle As System.Double  
  
instance.RotateAboutCenter(XAngle, YAngle)
```

C#

```
void RotateAboutCenter(  
    System.double XAngle,  
    System.double YAngle  
)
```

C++/CLI

```
void RotateAboutCenter(  
& System.double XAngle,  
& System.double YAngle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAngle
Rotation about the X axis
YAngle
Rotation about the Y axis

Visual Basic for Applications (VBA) Syntax

See [ModelView::RotateAboutCenter](#).

Example

[Change Wrap Feature Face \(C#\)](#)
[Change Wrap Feature Face \(VB.NET\)](#)
[Change Wrap Feature Face \(VBA\)](#)

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::RotateAboutAxis Method](#)
[IModelView::RotateAboutPoint Method](#)
[IModelView::StartDynamics Method](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

RotateAboutPoint Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : RotateAboutPoint Method (IModelView)

Rotates the model view about the specified point by the specified angles in the directions of the screen X and Y axes.

.NET Syntax

Visual Basic (Declaration)

```
Sub RotateAboutPoint( _  
    ByVal XAngle As System.Double, _  
    ByVal YAngle As System.Double, _  
    ByVal Ptx As System.Double, _  
    ByVal Pty As System.Double, _
```

```
ByVal Ptz As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim XAngle As System.Double  
Dim YAngle As System.Double  
Dim Ptx As System.Double  
Dim Pty As System.Double  
Dim Ptz As System.Double
```

```
instance.RotateAboutPoint(XAngle, YAngle, Ptx, Pty, Ptz)
```

C#

```
void RotateAboutPoint(  
    System.double XAngle,  
    System.double YAngle,  
    System.double PtX,  
    System.double Pty,  
    System.double Ptz  
)
```

C++/CLI

```
void RotateAboutPoint(  
& System.double XAngle,  
& System.double YAngle,  
& System.double PtX,  
& System.double Pty,  
& System.double Ptz  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAngle

Rotation about the screen X axis

YAngle

Rotation about the screen Y axis

Ptx

Center of rotation

Pty

Center of rotation

Ptz

Center of rotation

Visual Basic for Applications (VBA) Syntax

See [ModelView::RotateAboutPoint](#).

See Also

[IModelView Interface](#)

[IModelView Members](#)

[IModelView::RotateAboutAxis Method](#)

[IModelView::RotateAboutCenter Method](#)

[IModelView::RotateAboutPoint Method](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Scale Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Scale Property (IModelView)

Obsolete. Superseded by [IModelView::Scale2](#).

.NET Syntax

Visual Basic (Declaration)

Property Scale As System.Double

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double
```

```
instance.Scale = value
```

```
value = instance.Scale
```

C#

```
System.double Scale {get; set;}
```

C++/CLI

```
property System.double Scale {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::Scale](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Scale2 Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Scale2 Property (IModelView)

Gets and sets the model view scale factor.

.NET Syntax

Visual Basic (Declaration)

Property Scale2 As System.Double

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Double
```

```
instance.Scale2 = value
```

```
value = instance.Scale2
```

C#

```
System.double Scale2 {get; set;}
```

C++/CLI

```
property System.double Scale2 {  
    System.double get();  
    void set ( &    System.double value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Scale factor

Visual Basic for Applications (VBA) Syntax

See [ModelView::Scale2](#).

Example

[Get Scale Factor of Each Model View \(C++\)](#)
[Get Scale Factor of Each Model View \(C#\)](#)
[Get Scale Factor of Each Model View \(VB.NET\)](#)
[Get Scale Factor of Each Model View \(VBA\)](#)

Remarks

If you want to zoom in by a factor of 2, then you should get the current scale using this method, multiply the return value by 2, and then pass the value back into this method, which will increase the model view scale.

Using data returned from this property along with information from [IModelView::Orientation3](#) and [IModelView::Translation3](#) should be multiplied in this order: orientation > scale > transform.

To map your coordinates from 3D space to the screen, use [IModelView::Transform](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::!GetEyePoint Method](#)
[IModelView::!SetEyePoint Method](#)
[IModelView::SetEyePoint Method](#)

Availability

SOLIDWORKS 99, datecode 1999207

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetCameraByName Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : SetCameraByName Method (IModelView)

Sets the model view to that of the specified camera, if in camera view mode.

.NET Syntax

Visual Basic (Declaration)

```
Function SetCameraByName( _  
    ByVal Name As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Name As System.String  
Dim value As System.Boolean  
  
value = instance.SetCameraByName(Name)
```

C#

```
System.bool SetCameraByName(  
    System.string Name  
)
```

C++/CLI

```
System.bool SetCameraByName(  
& System.String^ Name  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name

Feature name of the camera to use to set the model view, if in camera view mode

Return Value

True if the camera is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::SetCameraByName](#).

Example

[Turn Cameras On and Off \(VBA\)](#)

Remarks

If you specify ...	Then...
A nonexistent camera when another camera view is active	Return value is false and the other camera view remains active
An empty string when another camera view is active	Return value is True and the camera view is turned off for this model view

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::Camera Property](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetEyePoint Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : SetEyePoint Method (IModelView)

Sets the eye position for perspective viewing.

.NET Syntax

Visual Basic (Declaration)

```
Function SetEyePoint( _  
    ByVal Eyept As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)

 Copy Code

```
Dim instance As IModelView  
Dim Eyept As System.Object  
Dim value As System.Boolean
```

```
value = instance.SetEyePoint(Eyept)
```

C#

```
System.bool SetEyePoint(  
    System.object Eyept  
)
```

C++/CLI

```
System.bool SetEyePoint(  
& System.Object^ Eyept  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Eyept

Array of 3 doubles describing the eye position in screen space

Return Value

True if the eye point is set successfully, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::SetEyePoint](#).

Remarks

The values are returned in pixels and are the location of the eye point relative to the screen-space origin (upper-left corner of window). The Z value, in pixels, reflects the distance from the eye point to the object center. These values will be affected by the view scale.

To use these values, you should apply them after you have obtained the current view orientation matrix using [IModelView::Transform](#).

The Z value typically are close to

[([IModelView::GetViewPlaneDistance](#) / ("Object Sizes Away")) * ("Object Sizes Away" + 0.5)]

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::AddPerspective Method](#)
[IModelView::GetEyePoint Method](#)
[IModelView::GetIsoPlaneDistance Method](#)
[IModelView::GetViewPlaneDistance Method](#)
[IModelView::HasPerspective Method](#)
[IModelView::IGetEyePoint Method](#)
[IModelView::ISetEyePoint Method](#)
[IModelView::RemovePerspective Method](#)
[IModelView::Scale2 Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetStereoSeparation Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : SetStereoSeparation Method (IModelView)

Obsolete and not superseded. Functionality no longer implemented.

.NET Syntax

Visual Basic (Declaration)

```
Function SetStereoSeparation( _  
    ByVal DfSeparation As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim DfSeparation As System.Object  
Dim value As System.Boolean  
  
value = instance.SetStereoSeparation(DfSeparation)
```

C#

```
System.bool SetStereoSeparation(  
    System.object DfSeparation  
)
```

C++/CLI

```
System.bool SetStereoSeparation(  
& System.Object^ DfSeparation  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DfSeparation

Array of 2 doubles representing the stereo magnitude and stereo parallax balance settings

Return Value

True if changing the stereo-value settings is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelView::SetStereoSeparation](#).

Remarks

This method only affects the model view's display if:

- Display window was set up to display stereoscopically
- Application is currently set to display stereoscopically.

[ISldWorks::EnableStereoDisplay](#) needs to be called to do these things.

There are two stereoscopic display values this method sets.

1. Settings[0] controls the magnitude of the stereoscopic effect by specifying the stereoscopic, virtual, camera separation. Appropriate values for this setting depend on the scale of the object, the distance from the virtual camera to the object center, the perspective field of view angle, and, perhaps, an adjustment factor that is available to the user.
2. Settings[1] controls the asymmetry of the projection frustums, thus, adjusting the parallax balance of the stereoscopic scene. It is desirable for any stereoscopic scene to comfortably balance the use of negative parallax (elements appearing to float in front of the display surface) and positive parallax (elements appearing to reside somewhere behind the display surface). Appropriate values for this setting should be arrived at experimentally.

If this method is not called, the default settings will equal 0.0, resulting in identical left-eye and right-eye renderings, and no visible stereo effect, even if stereoscopic display is currently enabled.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowMagnifyingGlass Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ShowMagnifyingGlass Method (IModelView)

Shows the Magnifying Glass tool at the specified coordinates.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowMagnifyingGlass( _  
    ByVal Ptx As System.Double, _  
    ByVal Pty As System.Double, _  
    ByVal Ptz As System.Double, _  
    ByVal ZoomFactor As System.Double, _  
    ByVal HideOnClick As System.Boolean, _
```

```
ByVal FollowMouse As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim Ptx As System.Double  
Dim Pty As System.Double  
Dim Ptz As System.Double  
Dim ZoomFactor As System.Double  
Dim HideOnClick As System.Boolean  
Dim FollowMouse As System.Boolean  
  
instance.ShowMagnifyingGlass(Ptx, Pty, Ptz, ZoomFactor, HideOnClick, FollowMouse)
```

C#

```
void ShowMagnifyingGlass(  
    System.double Ptx,  
    System.double Pty,  
    System.double Ptz,  
    System.double ZoomFactor,  
    System.bool HideOnClick,  
    System.bool FollowMouse  
)
```

C++/CLI

```
void ShowMagnifyingGlass(  
& System.double Ptx,  
& System.double Pty,  
& System.double Ptz,  
& System.double ZoomFactor,  
& System.bool HideOnClick,  
& System.bool FollowMouse  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Ptx
x coordinate
Pty
y coordinate
Ptz
z coordinate
ZoomFactor
Zoom factor
HideOnClick
True to hide the Magnifying Glass tool when the mouse is right-clicked, false to not
FollowMouse
True to have the Magnifying Glass tool follow the pointer, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelView.ShowMagnifyingGlass](#).

Example

Visual Basic for Applications (VBA)

```
' -----  
'  
' Preconditions: Model document is open.  
'  
' Postconditions: None  
'  
' -----  
Option Explicit  
  
Dim swApp As SldWorks.SldWorks  
Dim swModel As SldWorks.ModelDoc2
```

```

Dim swModelView As SldWorks.ModelView
Sub main()

Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swModelView = swModel.ActiveView

swModelView.ShowMagnifyingGlass -0.01928933522023, 0.004431675106825,
-0.001816629754713, 2, True, True
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.004
swModelView.MoveMagnifyingGlass -0.01928933522023, 0.004431675106825, -0.016
swModelView.HideMagnifyingGlass

End Sub

```

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::HideMagnifyingGlass Method](#)
[IModelView::MoveMagnifyingGlass Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

StartDynamics Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : StartDynamics Method (IModelView)

Provides faster rotation of model views.

.NET Syntax

Visual Basic (Declaration)

```
Sub StartDynamics()
```

Visual Basic (Usage)

```

Dim instance As IModelView
instance.StartDynamics()

```

C#

```
void StartDynamics()
```

C++/CLI

```
void StartDynamics();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::StartDynamics](#).

Example

[Dynamic View Rotation \(C++\)](#)

Remarks

After setting `swUserPreferenceIntegerValue_e.swLevelOfDetail` high enough, this method can be used at the beginning of dynamic model view changes to:

- change HLR or HLV mode to wireframe mode, which provides you with faster view rotation because the edges are not calculated during your view re-orientation.
- turn curved-surface bodies into polyhedra for faster view manipulations when in shaded mode.

Dynamic view motion remains in effect until you call [IModelView::StopDynamics](#).

If the SOLIDWORKS file is in view-only mode and is not displaying a shaded image, then you cannot perform any view rotations. Do not call any of the view rotation functions.

To determine if the file is in view-only mode and is not displaying a shaded image, use [IModelDoc2::IsOpenedViewOnly](#) and [IModelView::GetDisplayState](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::RotateAboutAxis Method](#)
[IModelView::RotateAboutCenter Method](#)
[IModelView::RotateAboutPoint Method](#)
[IModelView::DynamicMode Property](#)
[IModelView::HlrQuality Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

StopDynamics Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : StopDynamics Method (IModelView)

Ends dynamic model view motion.

.NET Syntax

Visual Basic (Declaration)

```
Sub StopDynamics()
```

Visual Basic (Usage)



```
Dim instance As IModelView
```

```
instance.StopDynamics()
```

C#

```
void StopDynamics()
```

C++/CLI

```
void StopDynamics();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::StopDynamics](#).

Example

[Dynamic View Rotation \(C++\)](#)

Remarks

Use this method with [IModelView::StartDynamics](#) at the end of dynamic view motion. This method returns the view to its previous HLR, HLV, or shaded mode and recalculates the display.

After calling this method, you must call [IModelDoc2::GraphicsRedraw2](#) or [IModelDoc2::WindowRedraw](#) to update the display for the end-user.

If the SOLIDWORKS file is in view-only mode and is not displaying a shaded image, then you cannot perform any view rotations. Do not call any of the view rotation functions.

To determine if the file is in view-only mode and is not displaying a shaded image, use [IModelDoc2::IsOpenedViewOnly](#) and [IModelView::GetDisplayState](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::RotateAboutAxis Method](#)
[IModelView::RotateAboutCenter Method](#)
[IModelView::RotateAboutPoint Method](#)
[IModelView::DynamicMode Property](#)
[IModelView::HlrQuality Property](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

SuppressWaitCursorDuringRedraw Property (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : SuppressWaitCursorDuringRedraw Property (IModelView)

Gets or sets the state of the wait cursor (also called the hourglass) while this model view is being redrawn.

.NET Syntax

Visual Basic (Declaration)

Property SuppressWaitCursorDuringRedraw As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Boolean

instance.SuppressWaitCursorDuringRedraw = value

value = instance.SuppressWaitCursorDuringRedraw
```

C#

```
System.bool SuppressWaitCursorDuringRedraw {get; set;}
```

C++/CLI

```
property System.bool SuppressWaitCursorDuringRedraw {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to enable the wait cursor, false to disable it

Visual Basic for Applications (VBA) Syntax

See [ModelView::SuppressWaitCursorDuringRedraw](#).

Example

[Disable Wait Cursor When Model View Redrawn \(VBA\)](#)

Remarks

When a model view is being redrawn, the wait cursor is normally displayed. This property allows you to suppress the wait cursor.

An add-in might find this property useful in situations where it is running a series of APIs, and one or more of those APIs cause the model view to redraw. By default, redrawing the model causes the wait cursor to flash on and off, which the end user may find disconcerting. By setting this property to false, the flashing wait cursor is suppressed. [IModelDoc::GraphicsRedraw2](#) is the most common API that causes a model view to redraw.

If your program disables this property, then it should re-enable it when the operation for which it was disabled completes. Otherwise, SOLIDWORKS will continue to *not* display the wait cursor. In other words, the expected use of this property is that you should disable it for a specific operation (or set of operations) in your code, and then immediately restore its previous behavior when that operation is completed.

Use of this property does not affect SOLIDWORKS performance.

See Also

[IModelView Interface](#)
[IModelView Members](#)
[IModelView::GraphicsRedraw Method](#)
[IModelView::IGraphicsRedraw Method](#)

Availability

SOLIDWORKS 2003 SP1, Revision Number 11.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

Transform Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Transform Property (IModelView)

Gets the model space to the model view plane transform.

NOTE: This property is a get-only property. Set is not implemented.

.NET Syntax

Visual Basic (Declaration)

Property Transform As MathTransform

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As MathTransform

instance.Transform = value
value = instance.Transform
```

C#

```
MathTransform Transform {get; set;}
```

C++/CLI

```
property MathTransform^ Transform {
    MathTransform^ get();
    void set ( & MathTransform^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[View plane transform](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::Transform](#).

Remarks

This method is typically used when you are grabbing the view handle using [IModelView::GetViewHWND](#) or [IModelView::GetViewHWNDx64](#) and drawing to the view. For example, if you had a point located at 2,2,2 in model space coordinates, then you could multiply it by this return value to determine where to draw in screen space coordinates. The result will be pixel values for the current view.

The screen space coordinate system has its origin in the upper-left corner of the current view with the X vector pointing to the right and the Y vector pointing down.

If the SOLIDWORKS file is in view-only mode and is not displaying a shaded image, then you cannot perform any view rotations. In this situation you should not call any of the view rotation APIs.

To determine if the file is in view-only mode and whether it is shaded or not, see [IModelDoc2::IsOpenedViewOnly](#) and [IModelView::GetDisplayState](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

[Send comments](#) on this topic.

TranslateBy Method (IModelView)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : TranslateBy Method (IModelView)

Translates the model view in the screen.

.NET Syntax

Visual Basic (Declaration)

```
Sub TranslateBy( _  
    ByVal X As System.Double, _  
    ByVal Y As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim X As System.Double  
Dim Y As System.Double  
  
instance.TranslateBy(X, Y)
```

C#

```
void TranslateBy(  
    System.double X,  
    System.double Y  
)
```

C++/CLI

```
void TranslateBy(  
& System.double X,  
& System.double Y  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

X
Translation in X direction, in meters and relative to X,Y axes of the graphics area
Y
Translation in Y direction, in meters and relative to Windows X,Y axes of the graphics area

Visual Basic for Applications (VBA) Syntax

See [ModelView::TranslateBy](#).

Example

[Set Viewports \(VB.NET\)](#)
[Set Viewports \(VBA\)](#)
[Set Viewports \(C#\)](#)

Remarks

This method lets you specify a vector by which to translate the current SOLIDWORKS graphics area. This vector is in meters and is relative to the X,Y axes of the graphics area. This vector has no relation to the SOLIDWORKS triad axis that is displayed in the graphics area. This method

is equivalent to the user-interface panning of the graphics area.

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Translation Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Translation Property (IModelView)

Obsolete. Superseded by [IModelView::Translation3](#).

.NET Syntax

Visual Basic (Declaration)

Property Translation As System.Object

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Object  
  
instance.Translation = value  
  
value = instance.Translation
```

C#

```
System.object Translation {get; set;}
```

C++/CLI

```
property System.Object^ Translation {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView::Translation](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

Translation2 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Translation2 Property (IModelView)

Obsolete. Superseded by [IModelView::Translation3](#).

.NET Syntax

Visual Basic (Declaration)

Property Translation2 As System.Object

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Object

instance.Translation2 = value

value = instance.Translation2
```

C#

```
System.Object Translation2 {get; set;}
```

C++/CLI

```
property System.Object^ Translation2 {
    System.Object^ get();
    void set ( &    System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelView.Translation2](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

Translation3 Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : Translation3 Property (IModelView)

Gets or sets the model view translation vector.

.NET Syntax

Visual Basic (Declaration)

```
Property Translation3 As MathVector
```

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As MathVector

instance.Translation3 = value

value = instance.Translation3
```

C#

```
MathVector Translation3 {get; set;}
```

C++/CLI

```
property MathVector^ Translation3 {
    MathVector^ get();
    void set ( &    MathVector^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

[Model view translation vector](#)

Visual Basic for Applications (VBA) Syntax

See [ModelView::Translation3](#).

Remarks

When modifying a model view transform, you must use [IModelView::Orientation3](#) and IModelView::Translation3. For example:

...

```
Dim xyzOrigin As SldWorks.MathPoint
dPoint(0) = 0#: dPoint(1) = 0#: dPoint(2) = 0#
Set xyzOrigin = swMathUtil.CreatePoint((dPoint))

Set swVectorZ = swVectorZ.Scale(-1)
Set swMathTrans = swMathUtil.ComposeTransform(swVectorX, swVectorY, swVectorZ,
xyzOrigin.ConvertToVector, 1#)
Set swMathTrans = swMathTrans.Inverse

Set swModelView = swModel.ActiveView
Set swViewTrans = swModelView.Orientation3
swModelView.Orientation3 = swMathTrans
```

```
Dim u As Double
u = swModelView.Scale2
Set swMathPoint = swVectorT.ConvertToPoint
Set swMathPoint = swMathPoint.Scale(-1 * u)
Set swMathPoint = swMathPoint.MultiplyTransform(swMathTrans)
swModelView.Translation3 = swMathPoint.ConvertToVector
...
```

Also, check the vectors by looking at the triad in the model view.

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2001Plus SP2, Revision Number 10.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

TurnBy Method (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : TurnBy Method (IModelView)

Rotates the camera by the specified angles about the camera's x and y axes.

.NET Syntax

Visual Basic (Declaration)

```
Sub TurnBy( _
    ByVal XAngle As System.Double, _
```

```
ByVal YAngle As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim XAngle As System.Double  
Dim YAngle As System.Double  
  
instance.TurnBy(XAngle, YAngle)
```

C#

```
void TurnBy(  
    System.double XAngle,  
    System.double YAngle  
)
```

C++/CLI

```
void TurnBy(  
& System.double XAngle,  
& System.double YAngle  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

XAngle

Angle by which to rotate the camera about its x axis

YAngle

Angle by which to rotate the camera about its y axis

Visual Basic for Applications (VBA) Syntax

See [ModelView::TurnBy](#).

Remarks

The model view must have an active camera.

See Also

[IModelView Interface](#)

[IModelView Members](#)

[IModelView::Camera Property](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpAxisFlyout Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : UpAxisFlyout Property (IModelView)

Gets or sets the up axis flyout flag.

.NET Syntax

Visual Basic (Declaration)

```
Property UpAxisFlyout As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Integer
```

```
instance.UpAxisFlyout = value  
value = instance.UpAxisFlyout
```

C#

```
System.int UpAxisFlyout {get; set;}
```

C++/CLI

```
property System.int UpAxisFlyout {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Up axis flyout option as defined by [swViewOrientationUpAxisFlyout](#) e

Visual Basic for Applications (VBA) Syntax

See [ModelView.UpAxisFlyout](#).

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2026 SP01, Revision Number 34.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpdateAllGraphicsLayers Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : UpdateAllGraphicsLayers Property (IModelView)

Gets or sets whether to update all graphic layers in any mode.

Visual Basic Syntax

Visual Basic (Declaration)

```
Property UpdateAllGraphicsLayers As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelView  
Dim value As System.Boolean
```

```
instance.UpdateAllGraphicsLayers = value  
value = instance.UpdateAllGraphicsLayers
```

C#

```
System.bool UpdateAllGraphicsLayers {get; set;}
```

C++/CLI

```
property System.bool UpdateAllGraphicsLayers {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to enable all graphic layers in any mode, false to not

 Visual Basic for Applications (VBA) Syntax

See [ModelView::UpdateAllGraphicsLayers](#).

 Remarks

SceneGraph organizes contents into layer 0 and 1. Layer 0 contains general geometry, and layer 1 contains sketch geometry. By default this property is set to false, which means only layer 1 is updated when in edit mode. When this property is set to true, then all layers are updated regardless of mode, including edit mode.

 See Also

[IModelView Interface](#)
[IModelView Members](#)

 Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

VisibilityViewTools Property (IModelView)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : VisibilityViewTools Property (IModelView)

Gets or sets the visibility of the Heads-up View Toolbar for this model view.

 .NET Syntax

Visual Basic (Declaration)

Property VisibilityViewTools As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelView
Dim value As System.Boolean

instance.VisibilityViewTools = value

value = instance.VisibilityViewTools
```

C#

```
System.bool VisibilityViewTools {get; set;}
```

C++/CLI

```
property System.bool VisibilityViewTools {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the Heads-up View toolbar is visible, false if not

 Visual Basic for Applications (VBA) Syntax

See [ModelView::VisibilityViewTools](#).

 Example

```
'-----
'
' Preconditions: Drawing document is open and
'               Heads-up View toolbar is visible.
'
' Postcondition: Heads-up View toolbar is hidden.
'
'-----
```

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModel As SldWorks.ModelDoc2
Dim swModelView As SldWorks.ModelView
Dim vRect as Variant
```

```
Sub main()
```

```
Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
Set swModelView = swModel.ActiveView
```

```
' Assume Heads-up View toolbar is visible (True)
Debug.Print "Heads-up View toolbar visible: " & swModelView.VisibilityViewTools
' Hide Heads-up View toolbar (False)
swModelView.VisibilityViewTools = False
Debug.Print "Hands-up View toolbar visible: " & swModelView.VisibilityViewTools
```

```
swModelView.GraphicsRedraw(vRect)
```

```
End Sub
```

Remarks

After setting this property to false, call [IModelView::GraphicsRedraw](#) or [IModelView::IGraphicsRedraw](#) to repaint the entire graphics window.

See Also

[IModelView Interface](#)
[IModelView Members](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

Gets or sets the Xor highlight state.

 .NET Syntax

Visual Basic (Declaration)

Property XorHighlight As System.Boolean

Visual Basic (Usage)

Copy Code

```
Dim instance As IModelView
Dim value As System.Boolean

instance.XorHighlight = value

value = instance.XorHighlight
```

C#

```
System.bool XorHighlight {get; set;}
```

C++/CLI

```
property System.bool XorHighlight {
    System.bool get();
    void set ( & System.bool value);
}
```

NOTE:

See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to Xor the highlight state, false to not

 Visual Basic for Applications (VBA) Syntax

See [ModelView.XorHighlight](#).

 See Also

[IModelView Interface](#)
[IModelView Members](#)

 Availability

SOLIDWORKS 2001 FCS, Revision Number 9.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ZoomByFactor Method (IModelView)

[See Also](#) [Example](#)

 

[SolidWorks.Interop.sldworks Namespace](#) > [IModelView Interface](#) : ZoomByFactor Method (IModelView)

Modifies the zoom factor for the model view.

 .NET Syntax

Visual Basic (Declaration)

```
Sub ZoomByFactor( _
    ByVal Factor As System.Double _
)
```

Visual Basic (Usage)

Copy Code

```
Dim instance As IModelView  
Dim Factor As System.Double  
  
instance.ZoomByFactor(Factor)
```

C#

```
void ZoomByFactor(  
    System.double Factor  
)
```

C++/CLI

```
void ZoomByFactor(  
    & System.double Factor  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Factor
Zoom factor

Visual Basic for Applications (VBA) Syntax

See [ModelView::ZoomByFactor](#).

Example

[Access Assembly \(C++\)](#)

Remarks

The zoom factor should be in the range $0.0 < \text{factor} < 2.0$. For factor greater than 1, the model view is zoomed in, for factor less than 1 the model view is zoomed out.

See Also

[IModelView Interface](#)
[IModelView Members](#)